Lab 4: Implementing *cat*

Objective: write a program that can concatenate and print files.

What to do?

- Write a C program *mycat* whose behavior resembles that of the system command cat. Unlike the standard cat, *mycat* will not work with the standard input. *mycat* must accept the following parameters:

```
mycat [-ens] file1 [file2 ...]
```

- *mycat* must receive at least one parameter, file1, whose content will be printed on the screen. If more than one file name is specified, the contents of all of the files will be printed.
- *mycat* must accept any combination of the following switches:
  - -e Display a dollar sign ('$') at the end of each line.
  - -n Number the output lines, starting at 1.
  - -s Squeeze (remove) all empty lines, causing the output to be single spaced. An empty line can contain one or more whitespace characters (e.g. space, tab, newline, carriage return)
- You may not use $fopen()$ and $fgets()$ to read the data from input files.
- Test *mycat* using the following text files: file1.txt, file2.txt, and file3.txt. Use the following test cases:

```
$ mycat file1.txt
$ mycat file1.txt file2.txt file3.txt
$ mycat -n file1.txt file2.txt
$ mycat -e file1.txt file2.txt
$ mycat -ne file1.txt file2.txt
$ mycat -es file1.txt file2.txt
$ mycat -ens file1.txt file2.txt file3.txt
```

What to submit

- A single C source code file with your work.
- One screenshot (in PNG or JPG format) showing the results of executing your program for the above test cases.